



Problem solving with similarity and Pythagoras' theorem

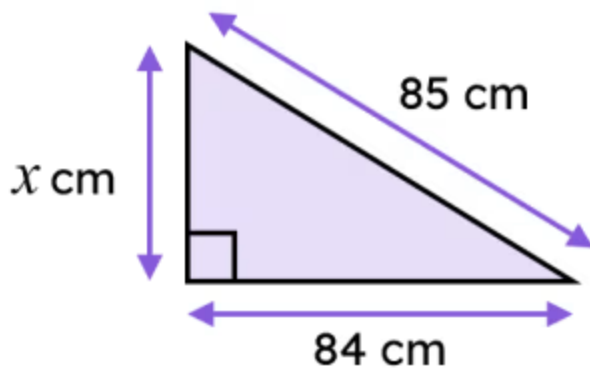
- 1 Alex rearranges a correct Pythagoras' theorem equation for a triangle. Which of these statements are correct? Tick **2** correct answers

$$65^2 - 25^2 = r^2$$

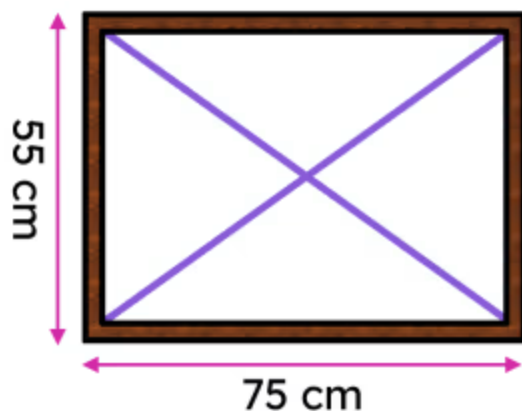
side P side Q side R

- ☐ The hypotenuse of the triangle is side P
- ☐ The hypotenuse of the triangle is side Q
- ☐ The hypotenuse of the triangle is side R
- ☐ Side R has length of 60 units.
- ☐ Side R has length of 69.64 units.

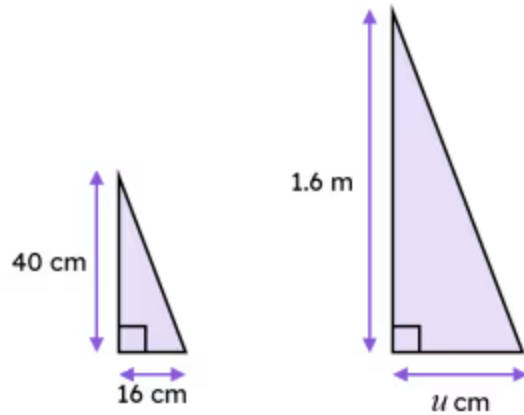
- 2 The perimeter of this right-angled triangle is _____ cm. Fill in the blank



- 3 Adhesive tape is placed along the two diagonals of this rectangular picture frame. How much adhesive tape is needed, rounded to the nearest cm? _____ cm of tape. Fill in the blank



- 4 Andeep and Sofia need to split a 3.6 metre roll of adhesive tape in the ratio 7 : 5. The length of the adhesive tape that Andeep receives is _____ centimetres. Fill in the blank
- 5 These two triangles are similar to each other. The side marked u cm is _____ cm long. Fill in the blank



- 6 An isosceles triangle has an angle of 22° . Which of these are possible sizes of one of the other angles? Tick 3 correct answers

- ☐ 11°
- ☐ 22°
- ☐ 45°
- ☐ 79°
- ☐ 136°

